



Cairo University  
Faculty of Engineering  
Computer Engineering Department  
CMPS303

Team 6

# Operating Systems

## Scheduler Operating System

### Project Report

### Team Number: 6

**Team Members:**

|                                |         |
|--------------------------------|---------|
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| Mennatullah Mohammad Abolfotoh | 1230266 |
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| Lojain Khaled Ramadan          | 1230086 |

## Data Structures Used:

- ❖ Linked List: used to read the processes from the input file and implement the priority and circular queues
- ❖ Priority Queue: used to implement HPF and SRTN with the enqueue modified based on an enum to use either priority or remaining time as the priority
- ❖ Circular Queue: used in the round robin to append the tail to the head

## Assumptions:

- Time Quantum = 2

## Algorithm Explanation and Results:

- ❖ SRTN: executes processes based on remaining time. The process with the shortest remaining time is pushed to the very front of the queue with its remaining time as the priority.
- ❖ HPF: executes processes based on given priority. The algorithm is pre-emptive so as to handle the dependency of one process on another of possibly lower priority.
- ❖ RR: executes processes in a circular queue fashion, with each process running for a particular time slice before it is pushed to the back of the queue. This happens until a process finishes.

## Workload Distribution and Time Taken:

- ➔ Mennatullah Mohammad Mahmoud: Process Generator and data structure + PCB struct implementation and report.
- ➔ Mariam Ramy Radwan: Shortest Remaining Time Next implementation and PCB handling, as well starting and switching between processes+IPC handling in scheduler
- ➔ Lojine Hany Mohamed: Pre-emptive Highest Priority First implementation + calculations.
- ➔ Lojain Khaled Ramadan: Process.c + Round Robin implementation +scheduler.log , scheduler.perf.

|            | Process Generator | Data Structures | SRTN  | HPF       | RR    | IPC Handling | Context switching | Process.c | Report  | Calc      |
|------------|-------------------|-----------------|-------|-----------|-------|--------------|-------------------|-----------|---------|-----------|
| Time taken | 1.5 days          | 6 hours         | 1 day | 6.5 hours | 15 hr | 6 hours      | 3 hours           | 30 min    | 2 hours | 1.5 hours |